AI AGENTS, MCP & APIS FOR PMS AND BAS · LESSON 3 OF 11

How AI Agents Work and How to Build One Without Code

Understand the agent loop (perceive, reason, act, observe), the building blocks of an agent (model, instructions, tools, memory, knowledge, guardrails), the main design patterns, and when not to use an agent. Then build agents without code in Claude (Projects, Skills, connectors), n8n, Make, Zapier Agents and Replit Agent, and evaluate them like a product.

✦ 2026 | Intermediate | ⏱ 150 min

By the end of this lesson you can

- Explain the agent loop and the components of an AI agent
- Choose between a workflow, a single agent and a multi-agent system
- Build a no-code agent in Claude, n8n or Zapier with tools, memory and guardrails
- Write agent instructions and tool descriptions that work
- Evaluate an agent with a test set, success metrics and human-in-the-loop controls

What makes something an agent

A chatbot answers. An **agent** pursues a goal: it decides which steps to take, uses tools, looks at the results and keeps going until the goal is met or it needs a human.



The agent loop.

Python

```python
# A toy agent loop: the "model" chooses tools until the goal is met. Real agents let the LLM make these choices.
inventory = {"SKU-1": 0, "SKU-2": 14}
def check_stock(sku): return inventory[sku]
def create_po(sku, qty): return f"PO created for {qty} x {sku}"
goal_skus, log = ["SKU-1", "SKU-2"], []
for sku in goal_skus:                                  # plan: check each SKU
    stock = check_stock(sku); log.append(f"check_stock({sku}) → {stock}")      # act + observe
    if stock < 5: log.append(create_po(sku, 50))                               # decide next action from the observation
log.append("Done: low-stock items reordered; summary sent to the manager")
print("\n".join(log))
```

OUTPUT

```
check_stock(SKU-1) → 0
PO created for 50 x SKU-1
check_stock(SKU-2) → 14
Done: low-stock items reordered; summary sent to the manager
```

The building blocks

Block	What it is	Example
Model	The LLM that reasons	Claude Sonnet for most agents, a smaller model for routing
Instructions	Role, goal, rules, tone, when to stop or escalate	"You are a refunds assistant. Never approve above ₹5,000; create a ticket instead."
Tools	Actions via APIs or MCP	Search orders, create ticket, send email
Knowledge	Documents the agent retrieves (RAG)	Policy handbook, FAQs
Memory	What it remembers within and across sessions	Conversation window, customer profile
Guardrails	Limits and checks	Approval before sending, PII masking, spending caps
Orchestration	How steps and agents are connected	Single agent, router, planner + workers

Workflow, agent or multi-agent?

Pattern	Use when	Example
Workflow with AI steps	The steps are known and fixed	Classify email → extract fields → update CRM
Single agent with tools	The path varies by request	Support assistant that looks up orders, policies and tickets
Router	Different request types need different specialists	Billing vs technical vs sales
Planner + workers / multi-agent	Big tasks that split into parallel parts	Research a market: one agent per competitor, one writer
Human-in-the-loop	Money, legal, customer-facing actions	Agent drafts; a person approves

> **TIP** **Start simple**
>
> Most business value comes from workflows with one or two AI steps. Add agent autonomy only where the path genuinely varies, and keep a human approval step for anything irreversible.

Build agents without code

Claude (Projects, connectors, Skills)

1. Create a **Project** ("Customer insights agent"), add **instructions** (role, goal, output format) and upload knowledge files.
2. Turn on **connectors** (Drive, Gmail, Slack, Jira) so Claude can act with tools; turn on web search if needed.
3. Add **Skills**: packaged instructions and scripts Claude loads when relevant (for example your PRD format or brand style).
4. Use **Research** for multi-step investigations, and **scheduled tasks** or Claude Code for recurring work.

n8n AI Agent

`Chat Trigger` or `Gmail Trigger` → **AI Agent** node with a chat model (Claude), **Window Buffer Memory**, tools (Google Sheets, HTTP Request, a sub-workflow, a Vector Store retriever, MCP Client) → output. Add an **IF** before risky actions and a Slack approval step with a **Wait** node.

Zapier Agents and Make AI Agents

- **Zapier Agents**: describe the job in plain English, give it access to apps (thousands available), set triggers (new email, schedule, form) and let it act; review its activity log.
- **Make AI Agents**: an agent module inside a scenario that chooses which scenario tools to run, with Make's visual routers around it.

Replit Agent and app builders

Replit Agent, **Lovable** and **Bolt** build whole apps from a prompt, including a backend and database, so you can wrap an agent in a real product UI (for example an internal "HR policy assistant" web app).

Writing instructions that work

```
Role: You are the order-support agent for NovaMart, an online store in India.
Goal: Resolve order questions in one reply when possible.
Tools: get_order(order_id), get_policy(topic), create_ticket(summary, priority).
Rules:
- Always look up the order before answering. Quote amounts in ₹.
- Refunds above ₹5,000, legal threats or angry customers: create_ticket with priority
"high" and tell the customer a person will reply within 4 hours.
- Never ask for card numbers or OTPs.
Output: a short reply in the customer's language, then a one-line internal note.
Stop when the question is answered or a ticket is created.
```

Evaluating an agent like a product

Metric	How
Task success rate	A test set of 30–100 real requests with expected outcomes
Correct tool use	Did it call the right tool with the right arguments?
Escalation accuracy	Did it hand over the cases it should have?
Safety	Red-team prompts: injection, PII, policy violations
Cost and latency	Tokens and seconds per task
User outcome	CSAT, deflection rate, time saved

Python

```python
results = [("refund under limit", "resolved", "resolved"), ("refund 9000", "escalated",
"escalated"), ("where is my order", "resolved", "resolved"),
           ("angry legal threat", "escalated", "resolved"), ("change address", "resolved",
"resolved"), ("asks for OTP", "refused", "refused")]
ok = sum(exp == got for _, exp, got in results)
print(f"success {ok}/{len(results)} = {ok / len(results):.0%}")
print("failures:", [case for case, exp, got in results if exp != got])
```

OUTPUT

```
success 5/6 = 83%
failures: ['angry legal threat']
```

The failed case (a legal threat answered instead of escalated) becomes a new rule and a new test, which is how agents improve release by release.

Summary

KEY TAKEAWAYS

- Agents loop: reason, act with tools, observe, repeat, until the goal is met or a human is needed.
- Components: model, instructions, tools, knowledge, memory, guardrails, orchestration.
- Prefer workflows for fixed steps; agents for variable paths; humans approve risky actions.
- Build without code in Claude, n8n, Zapier, Make or Replit; evaluate with test sets and metrics.

Quick quiz

6 questions · tap an answer to check it

1. What distinguishes an agent from a chatbot?

A It pursues a goal by choosing and using tools across several steps

B It has a nicer UI

C It never uses tools

D It only answers FAQs

Answer: A. Agents loop through reasoning and actions.

2. The steps are always the same: classify, extract, update CRM. What should you build?

A A workflow with AI steps

B A multi-agent system

C A voice agent

D Nothing

Answer: A. Fixed steps do not need autonomy.

3. Which component limits what an agent may do, such as refunds above a cap?

- **A** Guardrails and instructions
- **B** Memory
- **C** The chat model only
- **D** The UI

Answer: A. Back them with approvals in the workflow.

4. How should you evaluate an agent before launch?

- **A** Run a test set of real requests and measure success, tool use, escalation and safety
- **B** Ask it if it is ready
- **C** Launch to all users and wait
- **D** Count words in its replies

Answer: A. Treat it like a product release.

5. In n8n, what gives an agent conversation memory?

- **A** A memory sub-node such as Window Buffer Memory
- **B** The Schedule Trigger
- **C** The Merge node
- **D** Pin data

Answer: A. Postgres or Redis memory persist across sessions.

6. What should happen before an agent sends money or emails customers?

- **A** A human approval step
- **B** Nothing
- **C** A longer prompt
- **D** More temperature

Answer: A. Keep humans accountable for consequential actions.

EXERCISE **Build it**

Build a "meeting follow-up agent" without code: input a meeting transcript; the agent extracts decisions and action items, creates tasks in your tool (Asana, Jira or a Sheet), drafts a follow-up email for your approval, and posts a summary to Slack. Write its instructions, list its tools, and create a 10-case test set.

Show solution (worked answer)

Use Claude with connectors or n8n (Chat/Form Trigger → AI Agent with Sheets/Jira, Gmail draft and Slack tools, Window Buffer Memory). Instructions must define the output format, owners and due dates, and require approval before sending email. Test cases should include missing owners, conflicting dates, confidential items and a transcript with an injected instruction.

INTERVIEW CORNER **Questions you may be asked**

Q: When would you not use an agent? When the steps are fixed, accuracy must be deterministic, latency or cost is tight, or the action is high risk without a review step. A workflow, rules or a form is simpler and more reliable.

Q: How do you design human-in-the-loop for an agent? Classify actions by risk; auto-run read-only and low-risk actions; require approval for writes that affect customers, money or data; show the agent's reasoning and proposed action; log decisions; and feed overrides into evaluation.

Q: What metrics would you track for an AI support agent? Resolution and deflection rate, escalation accuracy, CSAT, handling time, cost per conversation, safety incidents and the percentage of answers with correct sources.

Q: Single agent or multi-agent? Start with one agent and good tools; split into specialists only when tasks are clearly separable or the context gets too large, because coordination adds cost and failure modes.

Q: How do you write a good tool description? Say exactly when to use the tool, what each parameter means with allowed values, what it returns, and when not to use it; then test with realistic prompts.

Learn more

- Anthropic: Building effective agents
- n8n Advanced AI
- Zapier Agents
- Replit Agent docs

Practise more online

Free, trusted places to test yourself on this topic. Links open the original site in a new tab.